35. Search Insert Position

Solved

Easy | Topics | Companies

Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [1,3,5,6], target = 5
Output: 2
```

Example 2:

```
Input: nums = [1,3,5,6], target = 2
Output: 1
```

Example 3:

```
Input: nums = [1,3,5,6], target = 7
Output: 4
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 <= nums[i] <= 10^4`
- `nums` contains **distinct** values sorted in **ascending** order.

Code

Java Auto

```java
class Solution {
    public int searchInsert(int[] nums, int target) {
        int left = 0;
        int right = nums.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (nums[mid] == target) {
                return mid;
            } else if (nums[mid] < target) {
                left = mid + 1;
            } else {
                right = mid - 1;
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums =
[1,3,5,6]

target =
5

Output

2

Description | Editorial | Solutions | Submissions

39. Combination Sum

Solved

Medium | Topics | Companies

Given an array of **distinct** integers `candidates` and a target integer `target`, return *a list of all* ***unique combinations*** *of* `candidates` *where the chosen numbers sum to* `target`. You may return the combinations in **any order**.

The **same** number may be chosen from `candidates` an **unlimited number of times**. Two combinations are unique if the frequency of at least one of the chosen numbers is different.

The test cases are generated such that the number of unique combinations that sum up to `target` is less than `150` combinations for the given input.

Example 1:

```
Input: candidates = [2,3,6,7], target = 7
Output: [[2,2,3],[7]]
Explanation:
2 and 3 are candidates, and 2 + 2 + 3 = 7. Note that 2 can be used multiple times.
7 is a candidate, and 7 = 7.
These are the only two combinations.
```

Example 2:

```
Input: candidates = [2,3,5], target = 8
Output: [[2,2,2,2],[2,3,3],[3,5]]
```

Example 3:

```
Input: candidates = [2], target = 1
Output: []
```


Code

Java | Auto

```java
import java.util.*;

class Solution {
    public List<List<Integer>> combinationSum(int[] candidates, int target) {
        List<List<Integer>> result = new ArrayList<>();
        backtrack(candidates, target, 0, new ArrayList<>(), result);
        return result;
    }

    private void backtrack(int[] candidates, int remaining, int start,
                           List<Integer> current, List<List<Integer>> result) {

        if (remaining == 0) {
            result.add(new ArrayList<>(current));
```

Saved

Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

candidates =

[2,3,6,7]

target =

7

Output

[[2,2,3],[7]]

40. Combination Sum II

Solved

Medium | Topics | Companies

Given a collection of candidate numbers (`candidates`) and a target number (`target`), find all unique combinations in `candidates` where the candidate numbers sum to `target`.

Each number in `candidates` may only be used **once** in the combination.

Note: The solution set must not contain duplicate combinations.

Example 1:

```
Input: candidates = [10,1,2,7,6,1,5], target = 8
Output:
[
[1,1,6],
[1,2,5],
[1,7],
[2,6]
]
```

Example 2:

```
Input: candidates = [2,5,2,1,2], target = 5
Output:
[
[1,2,2],
[5]
]
```

12.2K 225 71 Online

https://leetcode.com/problems/combination-sum?envType=problem-list-v2&envId=array

Code

Java | Auto

```java
import java.util.*;

class Solution {
    public List<List<Integer>> combinationSum2(int[] candidates, int target) {
        List<List<Integer>> result = new ArrayList<>();
        Arrays.sort(candidates);
        backtrack(candidates, target, 0, new ArrayList<>(), result);
        return result;
    }

    private void backtrack(int[] candidates, int remaining, int start,
                           List<Integer> current, List<List<Integer>> result) {

        if (remaining
```

Saved

Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 1 ms

Case 1 | Case 2

Input

candidates =

[10,1,2,7,6,1,5]

target =

8

Output

[[1,1,6],[1,2,5],[1,7],[2,6]]

45. Jump Game II

Solved

Medium | Topics | Companies

You are given a **0-indexed** array of integers `nums` of length `n`. You are initially positioned at index 0.

Each element `nums[i]` represents the maximum length of a forward jump from index `i`. In other words, if you are at index `i`, you can jump to any index `(i + j)` where:

- `0 <= j <= nums[i]` and
- `i + j < n`

Return *the minimum number of jumps to reach index* `n - 1`. The test cases are generated such that you can reach index `n - 1`.

Example 1:

```
Input: nums = [2,3,1,1,4]
Output: 2
Explanation: The minimum number of jumps to reach the last index is 2. Jump 1 step from index 0 to 1, then 3 steps to the last index.
```

Example 2:

```
Input: nums = [2,3,0,1,4]
Output: 2
```

Constraints:

- `1 <= nums.length <= 10^4`
- `0 <= nums[i] <= 1000`

Code

Java Auto

```java
class Solution {
    public int jump(int[] nums) {
        int jumps = 0;
        int currentEnd = 0;
        int farthest = 0;

        for (int i = 0; i < nums.length - 1; i++) {

            farthest = Math.max(farthest, i + nums[i]);

            if (i == currentEnd) {
                jumps++;
                currentEnd = farthest;
            }
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

nums =
[2,3,1,1,4]

Output

2

Expected

2

49. Group Anagrams

Solved

Medium | Topics | Companies

Given an array of strings `strs`, group the anagrams together. You can return the answer in **any order**.

Example 1:

> **Input:** `strs = ["eat","tea","tan","ate","nat","bat"]`
>
> **Output:** `[["bat"],["nat","tan"],["ate","eat","tea"]]`
>
> **Explanation:**
>
> - There is no string in strs that can be rearranged to form `"bat"`.
> - The strings `"nat"` and `"tan"` are anagrams as they can be rearranged to form each other.
> - The strings `"ate"`, `"eat"`, and `"tea"` are anagrams as they can be rearranged to form each other.

Example 2:

> **Input:** `strs = [""]`
>
> **Output:** `[[""]]`

Example 3:

> **Input:** `strs = ["a"]`
>
> **Output:** `[["a"]]`

21.9K 382 283 Online

Code

Java | Auto

```java
import java.util.*;

class Solution {
    public List<List<String>> groupAnagrams(String[] strs) {
        Map<String, List<String>> map = new HashMap<>();

        for (String str : strs) {
            char[] chars = str.toCharArray();
            Arrays.sort(chars);
            String key = new String(chars);

            map.putIfAbsent(key, new ArrayList<>());
            map.get(key).add(str);
        }
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

strs =
`["eat","tea","tan","ate","nat","bat"]`

Output

`[["eat","tea","ate"],["bat"],["tan","nat"]]`

Expected

`[["bat"],["nat","tan"],["ate","eat","tea"]]`

66. Plus One

Solved

Easy | Topics | Companies

You are given a **large integer** represented as an integer array `digits`, where each `digits[i]` is the `i^th` digit of the integer. The digits are ordered from most significant to least significant in left-to-right order. The large integer does not contain any leading `0`'s.

Increment the large integer by one and return *the resulting array of digits.*

Example 1:

```
Input: digits = [1,2,3]
Output: [1,2,4]
Explanation: The array represents the integer 123.
Incrementing by one gives 123 + 1 = 124.
Thus, the result should be [1,2,4].
```

Example 2:

```
Input: digits = [4,3,2,1]
Output: [4,3,2,2]
Explanation: The array represents the integer 4321.
Incrementing by one gives 4321 + 1 = 4322.
Thus, the result should be [4,3,2,2].
```

Example 3:

```
Input: digits = [9]
Output: [1,0]
Explanation: The array represents the integer 9.
Incrementing by one gives 9 + 1 = 10.
Thus, the result should be [1,0].
```


Code

Java Auto

```java
class Solution {
    public int[] plusOne(int[] digits) {

        int n = digits.length;

        for (int i = n - 1; i >= 0; i--) {

            if (digits[i] < 9) {
                digits[i]++;
                return digits;
            }

            digits[i] = 0;
        }
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

digits =
[1,2,3]

Output

[1,2,4]

Expected

[1,2,4]

73. Set Matrix Zeroes

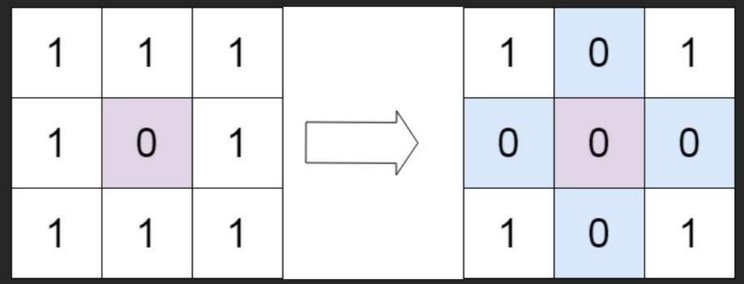
Solved

Medium | Topics | Companies | Hint

Given an `m x n` integer matrix `matrix`, if an element is `0`, set its entire row and column to `0`'s.

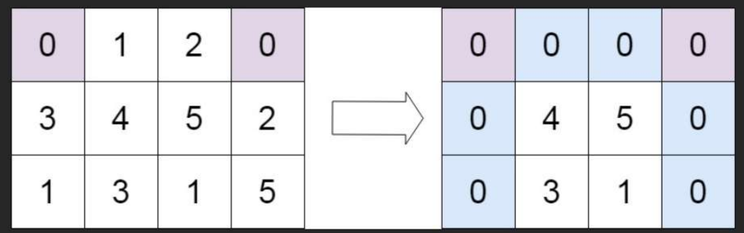
You must do it in place.

Example 1:




```
Input: matrix = [[1,1,1],[1,0,1],[1,1,1]]
Output: [[1,0,1],[0,0,0],[1,0,1]]
```

Example 2:




```
Input: matrix = [[0,1,2,0],[3,4,5,2],[1,3,1,5]]
```


Code

Java | Auto

```java
class Solution {
    public void setZeroes(int[][] matrix) {

        int m = matrix.length;
        int n = matrix[0].length;

        boolean fr = false, fc = false;

        for(int i = 0; i < m; i++)
            if(matrix[i][0] == 0)
                fc = true;

        for(int j = 0; j < n; j++)
            if(matrix[0][j] == 0)
```

Saved | Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

matrix =
[[1,1,1],[1,0,1],[1,1,1]]

Output

[[1,0,1],[0,0,0],[1,0,1]]

Expected

[[1,0,1],[0,0,0],[1,0,1]]

74. Search a 2D Matrix

Solved

Medium | Topics | Companies

You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `true` *if* `target` *is in* `matrix` *or* `false` *otherwise.*

You must write a solution in `O(log(m * n))` time complexity.

Example 1:

1	3	5	7
10	11	16	20
23	30	34	60

```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: true
```

Example 2:

Code

Java | Auto

```java
class Solution {
    public boolean searchMatrix(int[][] matrix, int target) {
        int i = 0;
        int j = matrix[0].length - 1;

        while (i < matrix.length && j >= 0) {
            if (matrix[i][j] == target)
                return true;
            else if (target > matrix[i][j])
                i++;
            else
                j--;
        }
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

matrix =
`[[1,3,5,7],[10,11,16,20],[23,30,34,60]]`

target =
`3`

Output

`true`

75. Sort Colors

Solved

Medium | Topics | Companies | Hint

Given an array `nums` with `n` objects colored red, white, or blue, sort them **in-place** so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers `0`, `1`, and `2` to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

Example 1:

```
Input: nums = [2,0,2,1,1,0]
Output: [0,0,1,1,2,2]
```

Example 2:

```
Input: nums = [2,0,1]
Output: [0,1,2]
```

Constraints:

- `n == nums.length`
- `1 <= n <= 300`
- `nums[i]` is either `0`, `1`, or `2`.

Follow up: Could you come up with a one-pass algorithm using only constant extra space?

21.4K 650 97 Online

Code

Java Auto

```java
class Solution {
    public void sortColors(int[] nums) {

        int low = 0, mid = 0, high = nums.length - 1;

        while (mid <= high) {

            if (nums[mid] == 0) {
                swap(nums, low, mid);
                low++;
                mid++;
            }
            else if (nums[mid] == 1) {
                mid...
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

nums =
[2,0,2,1,1,0]

Output

[0,0,1,1,2,2]

Expected

[0,0,1,1,2,2]

78. Subsets

Solved

Medium | Topics | Companies

Given an integer array `nums` of **unique** elements, return *all possible subsets (the power set)*.

The solution set **must not** contain duplicate subsets. Return the solution in **any order**.

Example 1:

```
Input: nums = [1,2,3]
Output: [[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]
```

Example 2:

```
Input: nums = [0]
Output: [[],[0]]
```

Constraints:

- `1 <= nums.length <= 10`
- `-10 <= nums[i] <= 10`
- All the numbers of `nums` are **unique**.

Seen this question in a real interview before? 1/5

Yes | No

19.1K 238 170 Online

Code

Java | Auto

```java
    private void backtrack(int[] nums, int start, List<Integer> current, List<List<Integer>> result) {

        result.add(new ArrayList<>(current));

        for (int i = start; i < nums.length; i++) {

            current.add(nums[i]);
            backtrack(nums, i + 1, current, result);
            current.remove(current.size() - 1); // backtrack
        }
    }
}
```

Saved Ln 22, Col 2

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

nums =
[1,2,3]

Output

[[],[1],[1,2],[1,2,3],[1,3],[2],[2,3],[3]]

Expected

[[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]

79. Word Search

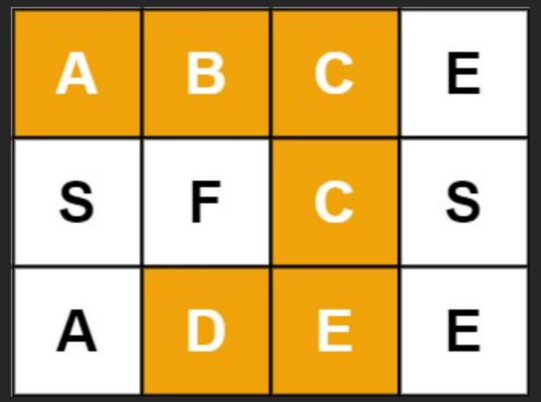
Solved

Medium | Topics | Companies

Given an `m x n` grid of characters `board` and a string `word`, return `true` *if* `word` *exists in the grid.*

The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

Example 1:




```
Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "ABCCED"
Output: true
```

Example 2:

Code

Java | Auto

```java
class Solution {
    public boolean exist(char[][] board, String word) {

        int m = board.length;
        int n = board[0].length;

        for (int i = 0; i < m; i++) {
            for (int j = 0; j < n; j++) {
                if (dfs(board, word, i, j, 0))
                    return true;
            }
        }

        return false;
```

Saved | Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

board =
```
[["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]]
```

word =
```
"ABCCED"
```

Output

```
true
```

18. 4Sum

Solved

Medium | Topics | Companies

Given an array `nums` of `n` integers, return *an array of all the* ***unique*** *quadruplets* `[nums[a], nums[b], nums[c], nums[d]]` such that:

- `0 <= a, b, c, d < n`
- `a`, `b`, `c`, and `d` are **distinct**.
- `nums[a] + nums[b] + nums[c] + nums[d] == target`

You may return the answer in **any order**.

Example 1:

```
Input: nums = [1,0,-1,0,-2,2], target = 0
Output: [[-2,-1,1,2],[-2,0,0,2],[-1,0,0,1]]
```

Example 2:

```
Input: nums = [2,2,2,2,2], target = 8
Output: [[2,2,2,2]]
```

Constraints:

- `1 <= nums.length <= 200`
- $-10^9 <= nums[i] <= 10^9$
- $-10^9 <= target <= 10^9$

Code

Java Auto

```java
import java.util.*;

class Solution {
    public List<List<Integer>> fourSum(int[] nums, int target) {

        List<List<Integer>> result = new ArrayList<>();
        int n = nums.length;

        Arrays.sort(nums);

        for (int i = 0; i < n - 3; i++) {

            if (i > 0 && nums[i] == nums[i - 1])
                continue;
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 2 ms

Case 1 Case 2

Input

nums =
[1,0,-1,0,-2,2]

target =
0

Output

[[-2,-1,1,2],[-2,0,0,2],[-1,0,0,1]]

34. Find First and Last Position of Element in Sorted Array

Solved

Medium | Topics | Companies

Given an array of integers `nums` sorted in non-decreasing order, find the starting and ending position of a given `target` value.

If `target` is not found in the array, return `[-1, -1]`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [5,7,7,8,8,10], target = 8
Output: [3,4]
```

Example 2:

```
Input: nums = [5,7,7,8,8,10], target = 6
Output: [-1,-1]
```

Example 3:

```
Input: nums = [], target = 0
Output: [-1,-1]
```

Constraints:

- `0 <= nums.length <= 10^5`
- `-10^9 <= nums[i] <= 10^9`
- `nums` is a non-decreasing array.

Code

Java Auto

```java
class Solution {
    public int[] searchRange(int[] nums, int target) {
        int first = -1, last = -1;
        for (int i = 0; i < nums.length; i++) {
            if (nums[i] == target) {
                if (first == -1) {
                    first = i;
                }
                last = i;
            }
        }
        return new int[]{first, last};
    }
}
```

Saved Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

nums =
[5,7,7,8,8,10]

target =
8

Output

[3,4]